বাংলাদেশ আর্মি ইন্টারন্যাশনাল ইউনিভার্সিটি অব সায়েন্স এন্ড টেকনোলজি (বিএআইইউএসটি) কুমিল্লা
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST) CUMILLA

Term Final Examination, Spring 2024

Department of Computer Science and Engineering
Level-2, Term-II
Course Code: MATH 221
Course Title: MATH IV (Probability and Statistics)
Credit Hour: 3.00

Notes:

Time : 2 hrs
Full Marks : 150

a. Each question carries 30 marks.
b. Answer any five (05) questions. **Either three from Part-A and two from Part-B, or two from Part-A and three from Part-B.**

Part-A

1. a. A coin is tossed three times. If X denotes the number of heads and Y denotes the number of tails in the last tosses, compute the joint probability distribution of X and Y. 10

   b. The Joint probability density of X and Y is given by: 20

   $$f(x, y) = \frac{x+y}{8}, \quad \text{for} \quad 0 \le x \le 2, 0 \le y \le 2$$
   $$= 0, \qquad \text{elsewhere}$$

   (i) Show that it is a joint density function. (ii) Find the marginal density function of X.
   (iii) Find P (X>Y) (iv) Find the conditional density function of Y.

2. a. Given the following joint density function of X and Y: 10

   $$f(x,y) = \begin{cases} 4xy, & 0 < x < 1, \quad 0 < y < 1 \\ 0, & \text{elsewhere} \end{cases}$$

   Observe that X and Y are independent.

   b. In a coin-tossing game, a man is promised to receive Tk.5 if he gets all heads or all tails when three coins are tossed and he pays off Tk.3 if either one or two heads appear. Identify how much is he expected to gain in the long run. 15

   c. The probability that a patient recovers from a delicate heart operation is 0.9. Estimate the probability that exactly five of the next seven patients undergoing this operation will survive? 05

3. a. The grade-point average score of 100 students of the Department of Computer Science of Dhaka University in their term-final examination was found to follow approximately a normal distribution with a mean of 2.1 and standard deviation 0.6. Find the number of students who are expected to have a score between 2.5 and 3.5. 12

   b. A life insurance company in Bangladesh offers to sell a T.25000 one-year term life insurance policy to a 25- year old man for a premium of Tk. 2500. According to Bangladesh life table, the probability of surviving one year for a 25-year old man is 0.97 and of his dying is 0.03. Compute the company's expected gain in the long run. 08

c. Let X and Y have the following distribution: 10

$$f(x,y) = x^2 + \frac{xy}{3}, \quad 0 \le x \le 1, 0 \le y \le 2$$
$$= 0, \qquad \text{elsewhere}$$

Show that $f(x,y)$ is a density function.

Part-B

1. a. A bag contains 10 balls of which 4 are black. If 3 balls are drawn at random without replacement, compute the probability distribution for the number of black balls drawn. 15

b. The Phillips Bangladesh manufactures electric bulbs that have a length of life that is normally with mean equal to 800 hours and standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours. 15

2. a. Construct a scatter diagram and fit the least-squares line to the given data using y as the dependent variable. 17

Grade on First Quiz (x)	6	5	8	8	7	6	10	4	9	7
Grade on Second Quiz (y)	8	7	7	10	5	8	10	6	8	6

b. In a sample survey of public opinion answer to the question 13
(i) Do you like to eat chocolate?
(ii) Are you in favor of the local chocolate bars? on sale of chocolate bars are tabulated below:

Question -2	Question -1		Total
	Yes	No	
Yes	56	31	87
No	18	6	24
Total	74	37	111

Determine your preference or not a local option for the sale of chocolate bars depends on individual choice. Use a 5% level of significance. ($\chi^2 = 3.841$)

3. a. Obtain a least-squares parabola having the form $y = a + bx + cx^2$ to the given data 17

x	1.2	1.8	3.1	4.9	5.7	7.1	8.6	9.8
y	4.5	5.9	7.0	7.8	7.2	6.8	4.5	2.7

b. In 200 tosses of a coin, 115 heads and 85 tails were observed. Determine the test of hypothesis that the coin is fair using a level of significance of 0.05. ($\chi^2_{0.05,1} = 3.84$) 13

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 2024

Department of Computer Science and Engineering

Level-2 Term-II

Course Code: CSE 225

Course Title: Numerical Methods

Credit Hour: 03

Exam Duration: 2 Hours

Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer ***any FIVE*** questions out of **SIX**.
c. Use different answer scripts for each section.

Part-A

1. a. Apply the Newton-Raphson method to find a root of the equation $f(x) = x^2 - 5$ starting from an initial guess $x_0 = 2$ up to 3 decimal places. 15

   b. Analyze the effect of different initial guesses on the convergence of one-point iteration for $g(x) = \sqrt{(1+x)}$. 15

2. a. Find the solution for $x = 1895$ using Newton's Forword and Backward Difference formula, taking into account the given table. Here, x is the year and $f(x)$ is the corresponding year's production. 30

x	1891	1901	1911	1921	1931
$f(x)$	46	66	81	93	101

3 a. Apply Simpson's $\frac{1}{3}$ rule to calculate $\int_0^1 e^x dx$. 15

   b. At $x = 3$, determine the Taylor series of $x^3 - 10x^2 + 6$. 15

Part-B

4. a. Consider an ordinary differential equation $\frac{dy}{dx} = x^2 + y^2$ where $y(1) = 1.2$ and $h = 0.05$. Produce $y(1.05)$ by fourth order Runge-Kutta method. 30

5 a. Applying Gauss-Seidel iteration method find the solution for the given equations: 30

$$27x + 6y - z = 85$$

$$x + y + 54z = 110$$

$$6x + 15y + 2z = 72$$

6 a. Use Gauss Quadrature formula, find the solution of $\int_0^1 x^2 e^{-x} dx$ for $n = 2$. 15

b. Calculate $\int_1^2 \frac{1}{x} dx$ for $n = 8$ by using Trapezoidal rule. 15

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Final Term Examination, Spring 2024

Department of Computer Science and Engineering

Level- 2 Term- II

Course Code: CSE 223

Course Title: Digital Logic Design

Credit Hour: 3

Exam Duration: 2 hours

Full Marks: 150

Notes:

a. Answer any five questions from the following.

b. Figure on the right of each question indicates marks for respective question.

PART-A

1. A flow rate sensing device used on a liquid transport pipeline functions as follows. The device provides a 5-bit output where all five bits are zero if the flow rate is less than 10 gallons per minute. The first bit is 1 if the flow rate is at least 10 gallons per minute; the first and second bits are 1 if the flow rate is at least 20 gallons per minute; the first, second, and third bits are 1 if the flow rate is at least 30 gallons per minute; and so on. The five bits, represented by the logical variables A, B, C, D, and E, are used as inputs to a device that provides two outputs Y and Z. **30**

   (a) Write an equation for the output Y if we want Y to be 1 iff the flow rate is less than 30 gallons per minute.

   (b) Write an equation for the output Z if we want Z to be 1 iff the flow rate is at least 20 gallons per minute but less than 50 gallons per minute. Design circuit for Y and Z

2. a. Draw the circuit diagram of two-bit parallel adder. Show the data flow in the circuit if A=01, B=11 **18**

   b. Explain different types of subtractor. **06**

   c. Draw the logic circuit of full adder. **06**

3. A metal cutting milling machine has two switches, any one will allow the cutter to run. The first switch is on the side of the machine and the second is a foot operated switch. However, the machine has two safety-switches (used as safety devices) if any of these is off the cutter will stop. The first safety-switch is on a guard, if this is opened the machine will stop. The second safety-switch is on a door which allows access to the moving mechanism of the milling machine. If this is opened the machine will stop. Derive the appropriate function for the problem. **30**

PART-B

4. a. Explain Clocked RS flip-flop with proper diagram 12

   b. Draw the circuit diagram of JK & D flip-flop using truth table. 18

5. a. Draw the circuit diagram for 8:3-line encoder. 15

   b. Design a 5:1 line multiplexer circuit. 15

6. Write down the equations for the following expressions: 4x7.5

   i) A company safe should be unlocked (A) only when Mr. Jones (B) is in the office or Mr. Evans (C) is in office, and only when the company is open for business (D), and only when the security (E) guard is present.

   ii) You should laugh at a joke if it is funny, it is in good taste, and it is not offensive to others, or if it is told in class by your professor (regardless of whether it is funny and in good taste) and it is not offensive to others.

   iii) You should wear your overshoes if you are outside in a heavy rain and you are wearing your new suede shoes, or if your mother tells you to.

   iv) The elevator door (D) should open if the elevator is stopped (S), it is level with the floor (F), and the timer (T) has not expired, or if the elevator is stopped, it is level with the floor, and a button is pressed (B).

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EXAMINATION CONFIDENTIAL

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BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST), CUMILLA

Term Final Examination, Spring 24
Department of Computer Science and Engineering
Level-2 Term-II
Course Code: CSE-221
Course Title: Computer Algorithms and Complexity
Credit Hour: 03
Exam Duration: 2 Hours
Full Marks: 150

Notes:

a. Figure on the right of each question indicate marks for respective question.
b. Answer *any FIVE* questions out of **SIX**.
c. Use different answer scripts for each section.

Part-A

1. a. You are given a knapsack with a weight capacity of 50 units. You have the following items, each with a weight and a value: **15**

Item	Weight (units)	Value (units)
A	10	60
B	20	100
C	30	120

Apply the Greedy knapsack algorithm to calculate the maximum value you can carry in the knapsack?

b. You are given the following set of jobs, each with a deadline and a profit. Find the maximum profit using Job Sequencing with Deadline algorithm. **15**

Job	Profit	Deadline
J_1	20	2
J_2	15	2
J_3	10	1
J_4	5	3
J_5	1	3

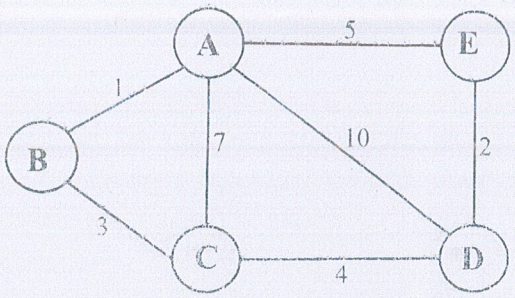
2. a. Given two sequences, X={A,B,C,D,G,H} and Y={A,E,D,F,H,R}. Determine the length and sequence of the longest subsequence that is common to both X and Y. **15**

b. You are given a bag with a weight capacity of **8** units. You have the following items, each with a weight and a value: **15**

Item	Weight (units)	Value (units)
A	2	1
B	3	2
C	4	5
D	5	6

Apply the 0/1 knapsack algorithm to find the maximum value and items you can carry in the knapsack.

3. **30**



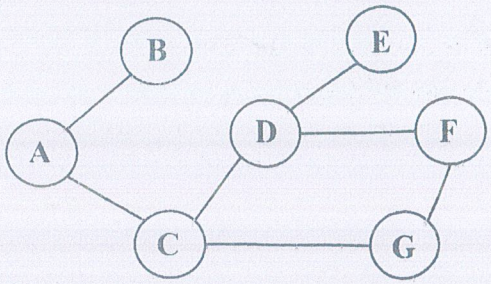
Illustrate the step-by-step execution of **Prim's and Kruskal's** algorithm on a given weighted graph, indicating how the MST is constructed.

Part-B

4. a. Briefly explain spanning tree and minimum spanning tree. **10**

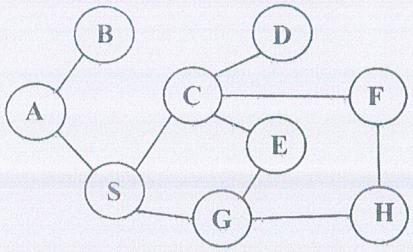
b. Apply Dijkstra's algorithm to find the shortest path from node A to node D in a weighted graph with the following edge weights: {(A, B, 4), (A, D, 3), (B, C, 2), (B, D, 5), (C, D, 1)}. Show the sequence of nodes visited and the final shortest path distance. **20**

5 a. 15



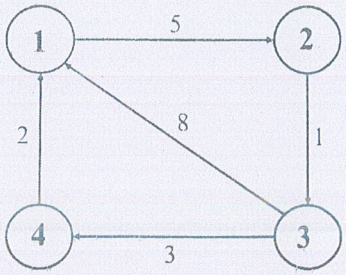
Explore each adjacent vertex in a Depth First Search (DFS) manner starting from vertex A. Provide the order in which the vertices are visited during the DFS traversal.

b. 15



Perform a Breadth First Search (BFS) starting from vertex A. List the order in which the vertices are visited during the DFS traversal.

6 a. 15



Implement the Floyd–Warshall algorithm to compute the shortest path between all pairs of vertices. Initialize the distance matrix using the adjacency matrix of the graph. Update the distance matrix iteratively using the Floyd–Warshall update rule.

b. Implement a backtracking algorithm to solve the N-Queen problem with state space tree where N=4. 15